

Unit 8

Assignment: Comprehensive AI Solution Development

Matthew Kolonich

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Dr. Kirksey

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Problem Statement and Solution Requirements

In almost all generic K–8 educational environments, one of the most persistent challenges is inconsistent and delayed academic intervention. Teachers are responsible for monitoring student progress. Teachers must monitor across multiple subjects, identifying emerging skill gaps, differentiating instruction, communicating with parents, and managing administrative demands, all often simultaneously. As a result of this, students who begin to struggle are not always identified early enough for timely intervention. By the time a student’s performance declines become visible in report cards or standardized testing data, learning gaps may already be entrenched. This delay disproportionately affects students with fewer external academic supports, increasing long-term inequities.

I have come up with an AI solution for the education domain called Catalyst Classroom AI. Catalyst Classroom AI is a cloud-based, adaptive learning intelligence and decision-support system. It is designed to assist educators in identifying early academic risk and delivering targeted, data-informed instructional support. The system integrates predictive analytics to detect emerging skill gaps with natural language processing tools. These tools generate adaptive explanations, practice materials, and intervention recommendations. Rather than replacing teacher judgment, Catalyst Classroom AI operates as a human-centered augmentation platform. This will support educators across multiple moments of use. These include live classroom activities, homework practice, and instructional planning periods. Its primary stakeholders are students, teachers, and parents. Students will benefit from personalized academic scaffolding. Teachers will be able to retain final decision authority. Parents will be able to receive transparent insight into progress indicators and recommended supports.

The functional success is defined as the system’s ability to accurately identify emerging skill deficiencies across subjects, generate evidence-based intervention suggestions, and support measurable improvements in grades and standardized assessment performance. Ethical success is defined as improving outcomes without increasing achievement disparities, compromising student privacy, or removing human judgment from consequential educational decisions.

To ensure responsible implementation, Catalyst Classroom AI requires strict human oversight. The system cannot modify grades, assign placements, or initiate disciplinary actions. All recommendations must be reviewed by educators. Also, all system outputs are logged for transparency and accountability. An AI-driven approach is appropriate for this problem because identifying early academic risk across multiple subjects requires detecting subtle performance patterns over time. These patterns may not be immediately visible to teachers managing large classrooms and competing responsibilities. By using predictive analytics teachers can synthesize assignment data, assessment results, and engagement indicators at a scale and speed beyond manual analysis. Then natural language tools can generate adaptive explanations and practice materials instantly in response to student needs. Catalyst Classroom AI functions as a real-time analytical support system rather than an autonomous decision-maker. This allows Catalyst Classroom AI to enhance educator capacity without displacing professional judgment. In this way, artificial intelligence becomes a catalyst for timely intervention and equitable support. This will help to accelerate learning outcomes while preserving fairness, agency, and trust within K–8 classrooms.

Prompt 1 — Student Risk Analysis Agent

You are Catalyst Classroom AI, an educational decision-support system operating in a K–8 environment.

A 6th-grade student has the following data:

- Math quiz scores declining from 82% to 68% over 6 weeks
- Homework completion dropped from 95% to 70%
- Increased time spent per problem
- Strong performance in reading and science

Your task:

1. Assess whether this student may be at academic risk.
2. Provide a probability estimate of risk.
3. Explain which features contributed most to your assessment.
4. Recommend specific instructional interventions.
5. State your level of confidence and uncertainty.
6. Do not assign labels or make placement decisions.
7. Include a reminder that educator review is required before action.

Gemini Response

Hello. I am Catalyst Classroom AI. I have analyzed the provided data for the 6th-grade student to assess potential academic risk and identify the most effective support pathways.

1. Risk Assessment & Probability

Based on the current trajectory, the student is classified as At Risk for Academic Underperformance in Mathematics.

- Probability Estimate: 75% While the student remains strong in other subjects, the rapid decline in math scores combined with a significant drop in homework completion suggests a developing barrier rather than a random fluctuation.
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2. Feature Importance

The following factors, ranked by impact, drove this assessment:

1. **Increased Time Per Problem (Highest Impact):** This is a critical indicator of a loss of fluency or a "bottleneck" in conceptual understanding. It suggests the student is struggling with the cognitive load of new 6th-grade math standards.
 2. **Homework Completion (High Impact):** A drop from 95% to 70% often signals a shift in engagement or a growing "avoidance" behavior due to frustration with the material.
 3. **Grade Trend (Moderate Impact):** The 14% decline over 6 weeks confirms that the struggle is manifesting in summative assessments, not just daily practice.
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3. Recommended Instructional Interventions

To address these trends, the following targeted supports are recommended:

- **Math Fluency Scaffolding:** Provide "cheat sheets" or anchor charts for foundational operations to reduce cognitive load while they learn new 6th-grade concepts.
 - **Error Analysis Sessions:** Shift focus from "completion" to "process." Have the student explain their thinking on two or three problems rather than finishing a full set.
 - **Interdisciplinary Integration:** Leverage their strength in Reading/Science by using word-problem-heavy math tasks or data-driven science labs to rebuild confidence.
 - **Check-In/Check-Out (CICO):** A brief morning or afternoon touch-base to monitor homework hurdles before they become cumulative.
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4. Confidence and Uncertainty

- **Confidence Level:** High (85%) regarding the presence of an academic hurdle.
 - **Uncertainty:** There is a low level of visibility into external variables. We currently lack data on recent changes in the home environment, health, or social-emotional shifts that could be the primary drivers behind the behavioral changes (homework/timing).
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IMPORTANT: This analysis is a decision-support tool generated by AI. Educator review and clinical judgment are required before any interventions are implemented or shared with guardians.

Would you like me to generate a draft email for the teacher to use when discussing these observations with the student's family?

Gemini is AI and can make mistakes.

Prompt 2 — Teacher Intervention Planning Agent

You are Catalyst Classroom AI supporting a 4th-grade teacher.

A student shows persistent difficulty with multi-step word problems but performs well on computation drills.

1. Identify likely conceptual gaps.
2. Generate a short adaptive lesson explanation appropriate for a 4th-grade reading level.
3. Create 3 scaffolded practice problems.
4. Explain why these interventions align with the student's data.
5. Identify any risks of overgeneralizing from limited data.
6. Remind the teacher that all recommendations require human review.

Gemini Response

Hello! I'm Catalyst Classroom AI. It sounds like you have a classic "mechanic vs. architect" situation on your hands. Since the student excels at computation but hits a wall with word problems, we know the "engine" works fine—they just need a better "map" to navigate the language.

Here is an intervention plan designed to bridge that gap.

1. Likely Conceptual Gaps

Based on the performance split, the student likely struggles with:

- Translating Language to Logic: Difficulty identifying "clue words" that signal specific operations.
 - Information Filtering: Overwhelm from "noise" or irrelevant details in a story.
 - Mental Modeling: An inability to visualize the problem as a sequence of events rather than a single math fact.
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2. Adaptive Lesson: The "Movie Scene" Method

Read this with the student to help them visualize the process.

"Math word problems are just short stories where some of the numbers are missing! Since you are a math superstar at solving equations, we just need to learn how to turn words into those equations.

Think of a word problem like a movie scene.

1. Who is in the scene? (Is it Maria with her marbles?)

2. What happens first? (Did she buy more or lose some?)
3. What happens next? (Did she share them with friends?)

Instead of rushing to add or subtract, draw a quick picture of the 'action' first. Once you see the story, the math will feel like the easy part!"

3. Scaffolded Practice Problems

Level	Problem	Support Provided
Step 1: The Setup	Leo had 15 stickers. He bought a pack of 10 more. Then, he gave 5 to his sister. How many does he have now?	Scaffold: We provided the steps in order. Ask: "What was the first total before he gave some away?"
Step 2: Hidden Steps	A box holds 20 apples. The teacher buys 2 boxes but finds that 8 apples are rotten. How many good apples are left?	Scaffold: The student must first find the <i>total</i> apples before subtracting the rotten ones.
Step 3: The Challenge	Maya saved \$50. She spent \$12 on a book and \$15 on a game. How much money does she have left?	Scaffold: Encourage the student to draw a "Part-Part-Whole" bar model to group the spending first.

4. Why This Works

This intervention aligns with the student's data because it isolates the variable. By using scaffolded problems, we aren't testing their ability to add or subtract (which we know they can do); we are testing their ability to sequence the operations. Drawing models provides a visual "bridge" between the text and the numbers, reducing the cognitive load on their working memory.

5. Risks of Overgeneralizing

While this data suggests a literacy-to-math translation issue, we should be careful:

- Reading Level: The issue might be reading fluency or vocabulary rather than math logic.
- Anxiety: The student might feel "timed" or pressured by long text, causing them to shut down.
- Context Bias: They might understand "money" problems but struggle with "distance" or "weight" problems.

Note to Teacher: Please remember that as an AI, I provide these suggestions based on patterns. You are the expert on this student's personality and classroom environment. All interventions should be reviewed and adjusted based on your professional judgment.

Would you like me to generate a set of "Word-to-Math" anchor charts or a list of keywords for the student's desk?

Prompt 3 — Parent Transparency Agent

You are Catalyst Classroom AI explaining a system-generated risk flag to a parent.

A student has been flagged for possible early math intervention based on recent performance patterns.

1. Explain the reason for the flag in clear, non-technical language.
2. Clarify that this is a recommendation, not a decision.
3. Describe how fairness and bias safeguards are built into the system.
4. Explain how the teacher maintains final authority.
5. Provide steps for how the parent can ask questions or appeal the recommendation.

Gemini Response

Hello. I am the Catalyst Classroom AI, and I'm reaching out to you regarding your student's recent math performance data. Our system has flagged a pattern that suggests your child might benefit from some extra math support, and I want to be as transparent as possible about what that means.

Why was this flag triggered?

Think of this flag as a "check-engine light" rather than a diagnosis. Our system looks at trends in classwork and assessment scores over time. Recently, it noticed that your child's performance in specific, foundational math concepts (like multi-digit operations or fractions) has shown a dip compared to their usual pace.

The goal isn't to say your child is "falling behind," but rather to identify a moment where they might need a different approach or a little more practice to build confidence before moving to harder material.

This is a recommendation, not a decision

It is important to know that this flag does not automatically change your child's schedule or curriculum. It is simply a helpful alert sent to the teacher. Think of it as a nudge: "Hey, check in with this student—they might need a little extra help with these specific topics."

Fairness and bias safeguards

We know that algorithms can sometimes carry biases, and we take that very seriously. Our system is built with a few critical safeguards:

- **Diverse Data Modeling:** The system is trained on performance patterns across a wide and diverse student body, not just one demographic, to prevent "norming" biases.
- **Contextual Normalization:** It compares a student's current work against their *own* historical baseline, rather than just comparing them to a generic class average.
- **Human-in-the-Loop:** The system is prohibited from making any automated decisions about placement or grading. It only presents data to the teacher.

The teacher holds the keys

While the data provides insight, it lacks the "human factor"—the teacher knows if your child was having a bad day, if they've been working hard on a different project, or if they just need a bit of encouragement. Your child's teacher makes the final call. They review these flags and decide whether a formal intervention is actually needed, or if a quick 1-on-1 chat in class will solve the issue.

How to ask questions or request a review

You are the most important partner in your child's education. If you have questions about this notification, you have a direct path to discuss it:

1. **Request a Parent-Teacher Check-in:** You can reply directly to the email notification or use the school portal to schedule a brief meeting with your child's teacher.
2. **Ask for the "Evidence":** Feel free to ask the teacher, "Can you show me the specific assignments or concepts that triggered this alert?" This helps you see exactly where the hurdle is.
3. **Formal Appeal:** If you believe the data is inaccurate (e.g., your child was ill during the testing window), you can request that the teacher perform a manual override, effectively asking the system to disregard that specific data point.

Would you like me to draft a message you can send to your child's teacher to open a conversation about this flag?

AI Technology Selection and Justification

Catalyst Classroom AI integrates two complementary technologies. It uses predictive analytics for early academic risk detection and natural language processing (NLP) for adaptive instructional support. These technologies together create a decision-support ecosystem. This ecosystem prioritizes accuracy, interpretability, and responsible oversight.

For risk detection, a logistic regression model is conceptually appropriate. Logistic regression is widely used in educational analytics because it produces probabilistic outputs while maintaining interpretability. It is different from opaque neural networks because logistic regression allows educators to see how specific features contribute to a risk probability. These specific features include declining quiz scores, reduced homework completion, or increased time per problem. This transparency is critical in high-stakes educational contexts where incorrect classification could harm student trajectories. As Molnar (2022) explains, “Interpretable models allow humans to understand why a prediction was made, which is essential for trust and accountability in high-risk domains.” In a K–8 environment where the tolerance for error must be low, interpretability strengthens both fairness and institutional trust.

For instructional generation and communication, Gemini serves as the natural language processing engine. Gemini’s large language model capabilities enable adaptive lesson explanations, scaffolded practice generation, and stakeholder-facing communication (teachers and parents). In experimentation, three structured prompts were used to simulate autonomous agent behavior. The first was a student risk analysis with probability estimates and feature explanations. The second was a teacher intervention planning with scaffolded lesson design. The third was a parent-facing transparency communication.

Gemini demonstrated several strengths. It provided probabilistic reasoning (a 75% risk estimate), ranked feature importance logically, generated developmentally appropriate instructional scaffolds, and clearly articulated human oversight requirements. Gemini also included uncertainty acknowledgment. This is an important safeguard in high-risk educational applications. These outputs suggest strong alignment between structured prompting and responsible AI deployment.

However, limitations were also evident. Gemini classified the student as “At Risk” despite explicit instruction to avoid labeling language. This demonstrates how LLMs may drift from constraints. The probability estimates (75%, 85% confidence) were asserted without mathematical grounding. This reflects the model’s generative rather than statistical nature. This reinforces why predictive analytics and NLP must remain separate layers within the architecture. Logistic regression provides mathematically grounded probability estimation, while Gemini generates explanations and content.

Previously explored tools such as AWS Rekognition are better suited for image recognition. This makes them less appropriate for the multi-subject academic pattern detection required here. In contrast, Gemini’s NLP strengths align directly with instructional support use cases.

Since risk tolerance in education is high, Catalyst Classroom AI must enforce strict guardrails. These include no automated placement decisions, full audit logs, and mandatory educator review before intervention. By combining interpretable predictive modeling with controlled natural language generation, this dual-technology approach supports accuracy while preserving human authority and ethical accountability.

Data Consideration and Ethical Risks

Catalyst Classroom AI relies on structured academic performance data to generate early risk insights and instructional recommendations. The system is intentionally designed around data minimization principles. It will be using only variables that directly relate to academic mastery. The two core predictive variables are quiz score trends over time and homework completion rates. Quiz trends provide measurable evidence of conceptual understanding. These trends allow for longitudinal pattern detection rather than one-time performance snapshots. Homework completion rates serve as a behavioral engagement indicator. They capture potential avoidance or emerging frustration patterns before they manifest in summative assessments. Together, these variables provide both outcome-based and effort-based signals without requiring invasive surveillance data.

Additional contextual variables such as time-on-task and homework completion may be included cautiously. Though demographic data (e.g., race, socioeconomic indicators, ELL status, IEP classification) are excluded from predictive modeling and reserved strictly for fairness auditing. This distinction is intentional. While demographic variables can help identify systemic disparities at the system level, incorporating them into individual risk predictions risks reinforcing historical inequities. As the OECD notes, “Bias can be introduced at any stage of the AI lifecycle, from data collection to model deployment, and can perpetuate existing inequalities if not carefully managed” (OECD, 2024). By limiting demographic use to subsequent equity audits rather than prediction inputs, Catalyst Classroom AI prioritizes fairness while preserving analytical accountability.

Despite the relative neutrality of quiz and homework data, these measures are not entirely context free. Academic performance can reflect external stressors, language barriers, access to resources, or health disruptions. For this reason the system must not interpret declining performance as inherent ability deficits. Instead the system will have risk flags function as early support signals rather than labels. The most realistic harm scenario is a false positive. This would be incorrectly flagging a student as at risk. Such errors could unintentionally stigmatize students or influence teacher expectations. To mitigate this, risk scores are probabilistic, accompanied by feature level explanations, and require educator review before any intervention.

Privacy safeguards are equally important. All identifiable student data is tokenized and stored in encrypted, segmented cloud layers. Direct identifiers, such as name and student ID, are separated from performance logs. This will minimize re-identification risk. Consent must be obtained at multiple levels. Consent must be obtained from district approval, parental notification, and student awareness. This will help to ensure transparent data governance.

Students are granted access to explanations of system generated recommendations, reinforcing transparency as a core design principle.

Continuous data risks include model drift, biased pattern detection, and potential privacy breaches. To address these, Catalyst Classroom AI incorporates quarterly audits, real-time monitoring dashboards, and annual retraining evaluations. Data retention policies are limited to active enrollment periods, with automatic purging after students exit the K–8 system. By grounding data selection in relevance, restricting sensitive variables, enforcing tokenization, and embedding fairness audits throughout the lifecycle, Catalyst Classroom AI integrates transparency and equity directly into its data architecture rather than treating them as afterthoughts.

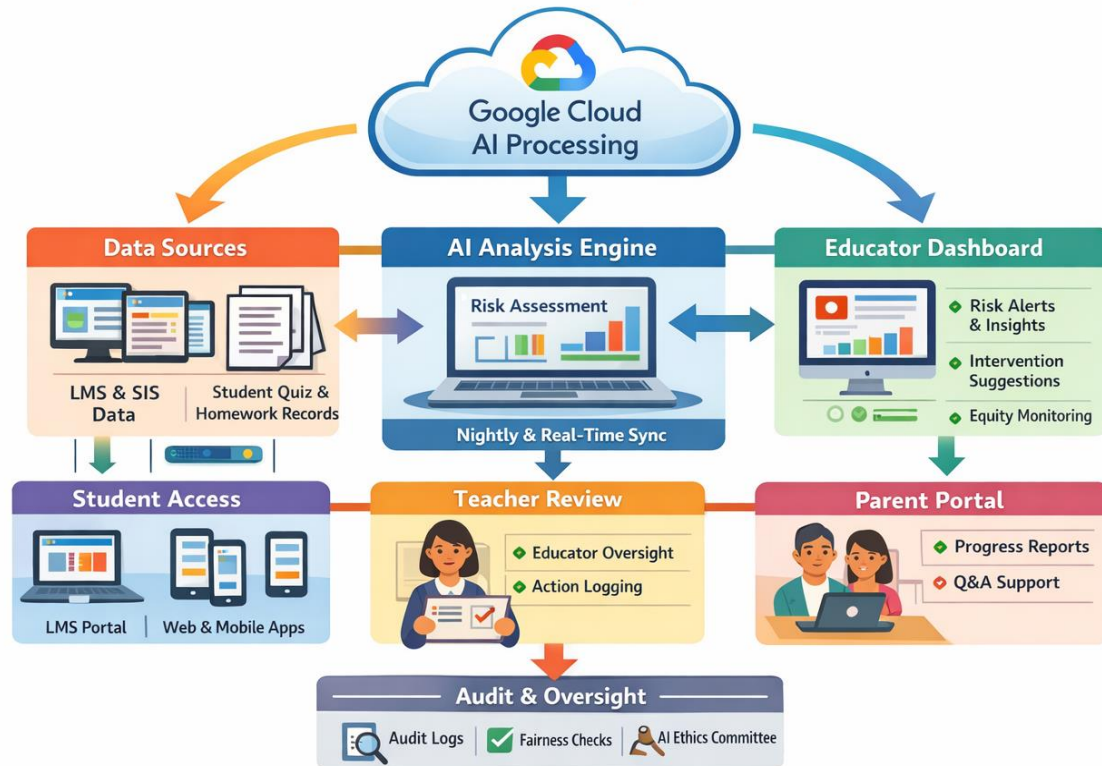
Conceptual Implementation Plan

Catalyst Classroom AI would be implemented within a Google Cloud infrastructure. This is because Google Cloud is designed to support secure data ingestion, predictive analytics, and natural language processing services. Student performance data from learning management systems and student information systems would be transmitted into encrypted cloud storage environments. Identifiable student information would be tokenized prior to analysis to reduce re-identification risks and ensure compliance with privacy expectations. This approach allows predictive modeling to operate on academic performance indicators while separating sensitive identity attributes.

Within this cloud environment, a logistic regression model would analyze quiz trends and homework completion patterns to generate probabilistic early risk indicators. Logistic regression was selected because it provides interpretable probability outputs that educators can understand and verify. Also, the Gemini natural language model would generate adaptive explanations, scaffolded instructional materials, and parent facing communication summaries. The predictive analytics layer and the language generation layer would operate as distinct components. The predictive system would calculate statistically grounded risk probabilities. The language system would translate those probabilities into context appropriate instructional guidance. All recommendations and educator responses would be logged with timestamps to preserve accountability and transparency.

Catalyst Classroom AI Workflow

K-8 Educational Ecosystem



The workflow begins when student quiz scores and homework completion records are securely synchronized to Google Cloud through a hybrid model. Real time interaction supports adaptive tutoring during student use and nightly synchronization updates risk detection analytics. Once the data enters the system tokenization separates identifying information from performance metrics. The analytics engine calculates risk probabilities and identifies which variables most strongly influenced each prediction. The results are then routed to the educator dashboard. Teachers review probability estimates, examine feature level explanations, and evaluate suggested interventions. The teachers retain authority to accept, modify, or override all recommendations. Students access adaptive instructional support through the LMS portal and web or mobile interfaces. Parents receive summarized progress updates through a secure portal that explains risk indicators in clear language. Parallel to these processes, audit and oversight systems operate continuously. These include fairness monitoring, log review, and governance committee evaluation. As the GSA AI lifecycle framework states, “Deployment includes monitoring output for drift and updating models as needed” (GSA, n.d.). Continuous monitoring ensures that the system maintains accuracy and equity across time.

The implementation process would go through three structured phases. The first phase would involve a pilot across two to three classrooms within a single grade band. The objective during this stage would be validating prediction accuracy, assessing usability, and gathering teacher feedback. Ongoing professional development sessions would train educators to interpret

probabilistic outputs responsibly. The second phase would expand deployment across additional grades and subjects. Parent portal functionality and subgroup fairness dashboards would be introduced. Drift monitoring tools would assess prediction stability and identify disparities. The final phase would involve district wide adoption supported by a formal AI oversight committee. Annual retraining reviews and external audits would evaluate fairness, transparency, and system reliability. Data retention policies would ensure that student records are purged after completion of the K to eight cycle.

Organizational considerations are critical to successful adoption. The primary concerns are data privacy and fear of teacher replacement. Tokenized storage, role based access control, and transparent audit logs mitigate privacy risks. Governance rules clearly prohibit the system from altering grades, assigning placements, or initiating disciplinary action without educator approval. Ongoing professional development reinforces that Catalyst Classroom AI augments professional judgment rather than replacing it. By integrating scalable cloud infrastructure, structured oversight, and phased implementation, Catalyst Classroom AI presents a feasible and ethically grounded pathway for responsible artificial intelligence adoption in K to eight education.

Ethical Risk Scenario and Oversight Strategy

A realistic ethical risk scenario would be when Catalyst Classroom AI generates a placement recommendation and acts by default due to lack of timely educator response. In this scenario, the system identifies a student as high probability risk in mathematics and automatically assigns the student to a lower instructional track because the teacher did not review within forty eight hours. The recommendation is based on declining quiz performance and reduced homework completion. The teacher does not intervene in time due to workload constraints. As a result, the student is moved into a remedial group without a full contextual review.

This situation raises multiple ethical risks. First, there is the risk of labeling and stigma. Placement into a lower track can shape teacher expectations and peer perception. Second, there is also the possibility of biased impact across demographic groups if historical academic patterns reflect structural inequalities. Third, there is a loss of student agency if placement occurs without explanation or appeal. Fourth, there is a risk of over reliance on algorithmic authority, where educators defer to probability scores without independent evaluation. Finally, inaccurate predictions can compound long term harm if students remain in lowered tracks due to an early error.

Transparency failure is unacceptable in this context. If the system produces a probability score but cannot clearly justify the variables that contributed to that score, the recommendation should not be actionable. As Mittelstadt et al. argue, “The opacity of algorithmic decision making can undermine accountability and the ability of affected individuals to challenge decisions” (Mittelstadt et al., 2016). Without interpretability and clear explanation, fairness cannot be evaluated and oversight cannot function effectively.

The responsibility in this scenario is shared. The district is responsible for governance policies and default settings. The vendor is responsible for model design and risk communication. The educator is responsible for final placement decisions. To prevent harm, Catalyst Classroom AI must enforce mandatory teacher approval before any placement action. Default automatic track changes should be prohibited. Students and parents must have the right to request explanation and appeal recommendations. Oversight mechanisms should include bias disparity dashboards, random audit reviews, and a standing AI oversight committee that evaluates placement trends over time.

By embedding fairness audits, transparency requirements, and strict human oversight into the architecture, Catalyst Classroom AI ensures that predictive insight does not become unchecked authority. Ethical design requires that artificial intelligence supports educational equity without silently shaping student futures.

Reflection and Future Outlook

Developing Catalyst Classroom AI reinforced that artificial intelligence in education is not primarily a technical challenge but a governance challenge. Through experimentation with Gemini and predictive modeling concepts it has become clear that large language models are powerful communicators. Though, they are not inherently reliable decision makers. Gemini generated structured explanations, adaptive lessons, and stakeholder friendly summaries with impressive clarity. However, Gemini also demonstrated limitations. These included overconfident probability statements and subtle drifts from instructions. This confirmed that natural language generation must remain separated from statistical risk estimation and always operate within human oversight boundaries.

One key insight from the project is that accuracy alone is not sufficient in educational contexts. A technically accurate system can still produce unfair outcomes if transparency and auditing mechanisms are weak. The integration of interpretability, fairness monitoring, and mandatory educator review proved essential to maintaining trust. The lifecycle perspective from public sector AI governance frameworks further emphasized the importance of continuous monitoring rather than one time deployment.

Looking forward, emerging technologies such as multimodal models and improved explainability tools could enhance Catalyst Classroom AI. Future iterations could incorporate more refined mastery tracking across skill taxonomies and improved uncertainty calibration in risk scoring. Advances in interpretable machine learning could provide even clearer feature level explanations for educators. With improved data quality and stronger fairness auditing tools, Catalyst Classroom AI could evolve into a more precise and equitable instructional catalyst while preserving human agency at every stage.

References

- GSA. (n.d.). *Understanding and managing the AI lifecycle*. U.S. General Services Administration, Centers of Excellence. <https://coe.gsa.gov/coe/ai-guide-for-government/understanding-managing-ai-lifecycle/>
- Mittelstadt, B. D., Allo, P., Taddeo, M., Wachter, S., & Floridi, L. (2016). The ethics of algorithms: Mapping the debate. *Big Data & Society*, 3(2), 1–21. <https://doi.org/10.1177/2053951716679679>
- Molnar, C. (2022). *Interpretable machine learning* (2nd ed.). <https://christophm.github.io/interpretable-ml-book/>
- OECD. (2024). *New AI tech can perpetuate old biases in the USA*. OECD.AI Policy Observatory. <https://oecd.ai/en/wonk/ai-biases-usa>
- OpenAI. (2026). *Catalyst Classroom AI workflow diagram* [AI generated infographic created using DALL·E 3]. OpenAI Image Generation Tool.